

IT AT AIRPORTS: Today And Tomorrow



Deepak Holan is an associate professor at School of Management Sciences, Apeejay Stylo University

"The new Terminal 2 isn't just one building but a complex machine comprised of 118,000 individual assets that need to work in harmony to deliver a good experience"—John Holland-Kaye, CEO, Heathrow Airport

Amsterdam's Schiphol and London Heathrow airports have installed self-service biometric passport gates that use facial recognition technology. Japan's All Nippon Airways, on the other hand, makes smart tablets available to travellers so that they can check in quickly, navigate the airport, and receive real-time messages about their flights. Information technology is quickly transforming aviation and touching every aspect of air travel from the point of departure to arrival at destination. The airports of tomorrow are going to be a very new and exciting experience and are expected to become one of the more pleasing parts of the journey, rather than one of the most boring.

The travel experience will begin the moment you make your reservation. For example, on arrival at the airport, you can choose how to check in: auto check-in when entering the terminal, use a self-service electronic kiosk or stand at a conventional desk-based check-in. To check-in bags, self-tagging stations will allow you to efficiently manage

your luggage. If you have no bags, you go straight for security check where identity sensors will confirm your identity.

Your boarding pass, identification and biometrics will allow you total control of passage through the airport, all the way to aircraft boarding. The key to this automated travel vision is biometrics, in the form of iris scan, fingerprint, facial recognition, etc. Digital technology will eliminate holdups in check-in, bag drop, security and boarding by doing away with the most time-consuming element—human interaction. Biometrics in every phase of the process will help airports to remain vigilant with passengers. For example, on inbound international flights, the passenger verification process will complete even before the aircraft lands.

Soon your face will be your passport. Australia has developed the technology to introduce fingerprints, iris or facial recognition at major air and seaports to replace passports for passenger identification by 2020. This will help to improve security by making it easier to detect threats. The system will also avoid paperwork or manual processing for majority of the passengers and make it easier and quicker for people going in and out of airports.

Integration and wireless innovations

Fully integrated, end-to-end solutions using information and wireless technology will eliminate the use of large amounts of cables, leading to lower costs, greater efficiency, greater mobility and geo-location, and a greatly enhanced passenger experience. For example, using wireless beacon technology and a simple smart phone app, individual or group messages can be sent to inform only the targeted passengers of relevant gate changes, or ground handlers can be apprised of a weather forecast change. These systems will identify the closest ground handler to secure vehicles nearby.

Self-service biometric passport gate being used by a passenger at Heathrow Airport (Image courtesy: www.futuretravelexperience.com)

